

Catalonia

Catalan Competition Authority

For over a year, the Catalan Competition Authority ('ACCO') has been developing a computer tool named ERICCA, which uses big data and artificial intelligence to assist economists and lawyers working in this organization.⁶ Particularly, this assistance includes both, simplified access to public procurement data in the context of bid rigging investigations arising from complaints and inquiries made to this organization, as well as a tool that facilitates the proactive identification of public tenders where it is most likely that participating companies have established collusive agreements for the submission of their proposals.

To access information related to Catalan public tenders, in mid-2022 ACCO signed a collaboration agreement with the Department of Economy and Finance of the Generalitat de Catalunya, the data owner of the Public Contracts Register (RPC).⁷ The RPC is the electronic register containing the basic data of the public contracts awarded by the contracting bodies of the Catalan government and the Local governments, as well as the rest of entities subject to the public sector contracts law.

This agreement opened access to a broader range of tender-specific information, since information gathered from RPC's website before the agreement was less complete.⁸ It made it possible to access details from many tenders, including not just the name of the winning company, but also the names of all other participating companies and their respective bids. This extensive information is vital for conducting analyses that aim to identify tenders where participants are suspected of having had a coordinated behavior.

Currently, the tool has already been able to identify several groups of companies that, in certain tenders, could have reached anti-competitive agreements, which would have allowed them to win the tenders without incurring in competitive

⁶ The Catalan Competition Authority (ACCO) is an organization whose mission is to ensure the correct competitive functioning of markets in all productive sectors of the economy in Catalonia. Autoritat Catalana de la Competència, https://acco.gencat.cat/ca/l_acco/que_es_i_que_fa_l_acco/index.html (last accessed June 7, 2023).

⁷ Contractació pública, <https://contractacio.gencat.cat/ca/contractacio-electronica/registres-electronics/rpc/> (last accessed June 7, 2023).

⁸ Previously, information was obtained using web scraping techniques from the Public Procurement Services Platform. Contractació pública, <https://contractaciopublica.cat/ca/inici> (last accessed June 7, 2023).

discounts, thus potentially obtaining supra-competitive profits. These tenders and companies involved in them are being analyzed thoroughly by ACCO.

However, we have observed that, despite the large number of tenders collected in the currently used database, about 250,000, corresponding to tenders carried out between 2010 and 2023, it is a relatively small sample compared to the total amount of tenders carried out during this period.⁹

This limitation on the data representatively encouraged us to implement our first improvement in the tool. We decided to complement our current database with data from the OPEN DATA portal, which provides information on nearly 2.000,000 tenders, significantly expanding our data resources.¹⁰ The OPEN DATA portal includes sets of data produced or collected by public bodies that public administrations make available to citizens so that they can use them freely in a simple and convenient way.¹¹ Although this database does not include the participating companies in each tender, the name of the winning company and the amount of the award are indicated, information that can be very useful, among other things, to be more efficient in identifying collusive rotation agreements (e.g. possible temporary distributions or tendering body distributions).

Secondly, we have continued to advance in the incorporation of additional statistical parameters, identified in the specialized literature, which should facilitate the detection of anti-competitive behaviors. Currently, for each group of companies that the artificial intelligence algorithm clusters, the software:

- Calculates the individual discount of each company and the group as a whole and shows the average coefficient of variation of these variables (statistic that measures dispersion).
- Indicates the number of times that companies in the same group have made exactly the same economic offer (*i.e.*, suspicious circumstance of a possible agreement).
- It also calculates the percentage of tenders won by a specific company, taking into account the name of companies participating in the tenders.

⁹ Considering each lot of the same tender as an independent tender.

¹⁰ Dades Obertes Catalunya, <https://analisi.transparenciacatalunya.cat/Sector-P-blic/Contractaci-de-Catalunya/hb6v-jcbf> (last accessed June 7, 2023).

¹¹ Dades Obertes Catalunya, https://governobert.gencat.cat/ca/dades_obertes/dades-obertes/que-son-dades-obertes/ (last accessed June 7, 2023).

This helps identify potential 'accompanying offers' that may have been designed to increase the likelihood of a particular company winning.

Another criterion for spotting anti-competitive behavior that we are adding to the algorithm involves segmenting by economic sector (CPV) and geographic location.¹² This will help us identify potential agreements to divide territories among companies.

Thirdly, we are investing in enhancing the usability and designing a user-friendly interface. For instance, the tool currently can display all tenders from the same group of companies, which can be sorted by various criteria, such as the date of award or the tendering body, allowing users to visually check for any rotation among the winners. To make this detection easier, we are introducing indices (with values ranging from -1 to 1) that calculate whether there's a rotation in the awards among the group's companies.

Moreover, the tool is being improved in order to allow users to easily download the information consulted in CSV format, which can be included in the working documents and reports or can be imported to other computer programs to complement and enrich the analysis with other software.

Lastly, it is worth mentioning that several institutions and public organizations have reached out to ACCO, interested in understanding how the tool was developed and in exploring potential collaborations.

¹² The CPV nomenclature (Common Procurement Vocabulary) is a system of identification and categorization of all economic activities that can be contracted through tendering or public competition in the European Union.